

Tech Stack

Component	Technology
Frontend	React + Vite
Backend	Python + FastAPI
Cryptography	Open Quantum Safe / liboqs
PQ Key Exchange	ML-KEM-768
PQ Digital Signature	ML-DSA-65
Hashing	SHA-3-256
Randomness	OS CSPRNG
Document encryption	AES-256-GCM
Watermark	DCT/DWT-based invisible watermark
Ledger	Hyperledger Fabric or custom permissioned blockchain
Database	PostgreSQL / SQLite
Identity	Local PKI + recipient key registry
Document processing	Python + OpenCV/Pillow/PyMuPDF
Frontend ↔ backend	REST API
Containerization	Docker
Deployment	Local LAN / completely offline
OS	Windows/Linux
Cloud	None
Public blockchain	None